
EXERCISE

Lecture 5: Sealed Bid Auctions 2

AI508 — Algorithmic Game Theory

Simon Holm
Fall - 2026



Contents

1 Third Price Auction 3

2 Sponsored Search (simplified) 3

3 Mechanism 4

1 Third Price Auction

Highest bidder gets the item, but pays the amount of equal to the third highest bidder

$$B_i \stackrel{\text{def}}{=} \max_{j \neq i} b_j$$

where

$$x_i(b_i) = \begin{cases} 1 & \text{if } b_i > B_i \\ 0 & \text{if } b_i < B_i \end{cases}$$

Definition: Welfare Maximization in sealed bid auctions

Given 1 item, $x_i \in \{0, 1\}$ and x is a valid allocation

$$\max_x \sum_{i=1}^n v_i x_i$$

2 Sponsored Search (simplified)

Given $|J|$ ad slots and n bidders. Then

$$1 \geq \alpha_1 \geq \dots \geq \alpha_{|J|} \geq 0$$

Each bidder has only 1 bid with value v_i (value per click)

The i^{th} bidder gets allocated j^{th} slot

$$x_i \in \{0, \alpha_1, \dots, \alpha_{|J|}\}$$

Example:

given that $\alpha_1 = 1, \alpha_2 = 1/2$, and that other bids are 5 and 8. Then

$$x_{i(b_i)} = \begin{cases} 0 & \text{if } b_i < 5 \\ 1/2 & \text{if } 5 < b_i < 8 \\ 1 & \text{if } b_i > 8 \end{cases}$$

Here one can either get the best slot 🤩, second best slot 😊 or no slot at all 😞.

3 Mechanism

Given the utility function

$$u_i = v_i x_i - p_i$$

Definition: Mechanism rules

A mechanism in this sense is function that defines allocation rule x and payment rules p .

$$b \mapsto (x(b), p(b))$$

A mechanism is dominant strategy incentive compatible (DSIC) if

$$\underbrace{v_i x_i(v_i, b_{-i}) - p_i(v_i, b_{-i})}_{\text{truthful bidding}} \geq v_i x_i(b_i, b_{-i}) - p_i(b_i, b_{-i}) \quad \forall v_i, b_i, b_{-i}$$

Definition: Monotone allocation

For each player i , an allocation is monotone if for $b_i > b'_i$, it holds that

$$x_i(b_i, b_{-i}) \geq x_i(b'_i, b_{-i})$$

If b_i increases you will have the *same or better* allocation

Myerson's Lemma

For (x, p) DSIC $\iff X$ is a monotone allocation

Then derived from Myerson's Lemma is:

$$p_i(b_i, b_{-i}) = b_i x_i(b_i, b_{-i}) - \int_0^{b_i} x_i(z, b_{-i}) dz$$

Note that if you change the *true* value DSIC still holds given that

$$zx(z) - p(z) \geq zx(y) - p(y)$$

$$yx(y) - p(y) \geq yx(z) - p(z)$$